

frame buffer before being transported to the display medium so that you and I can enjoy it.

Previously, graphics cards had separate shaders for each task that had to be performed. So there were pixel shaders, vertex shaders and geometry shaders. Given how complex and variable a 3D scene can be, not every shader ends up being used at all times. The introduction of unified shaders made this problem go away. Each unified shader can now perform the task of the individual types of shaders that used to exist previously. Also, since these shader units are programmable, the graphics card became sort of a general purpose accelerator and thus, began the age of GPGPU (General Purpose Computing on Graphics Processing Units) computing. Let's now see how NVIDIA and AMD differ from each other at a microarchitecture level.

[JARGON BUSTER]

Etching

Etching is a process used in microfabrication where a chemical (usually hydrofluoric acid) is used over a mask to remove away layers from the surface of a silicon wafer.

Fabrication

This refers to the entire process of producing an integrated circuit, and involves a series of chemical and photolithographic operations on a semiconducting wafer or substrate.

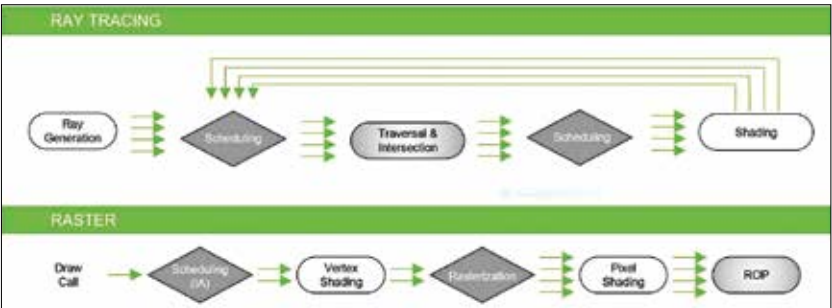
NVIDIA

NVIDIA's latest microarchitecture is called Turing and it has a simple hierarchical organisation of all the unified shaders. NVIDIA calls its unified shaders, CUDA cores and they're grouped into Streaming Multiprocessors (SM). Each SM consists of 64 CUDA cores, 8 Tensor Cores, a 256 KB register, 4 Texture Units and 96 KB of L1 cache memory that is shared among multiple SMs. 2 SMs, are in turn, grouped into TPCs (Texture Processing Cluster) and then 6 TPCs are grouped along with a raster engine into what's called a GPC (Graphics Processing Cluster). With the Turing architecture, NVIDIA has also introduced a new type of shader called RT (Ray Tracing) Cores to better handle the real-time ray-tracing feature that was introduced recently. Since graphics cards have also become useful for accelerating general compute, there are Tensor Cores in each GPU which help with AI workloads. All of this has a massive shared L2 cache pool. And all raster engines plug into memory controllers which help move the data to the VRAM.

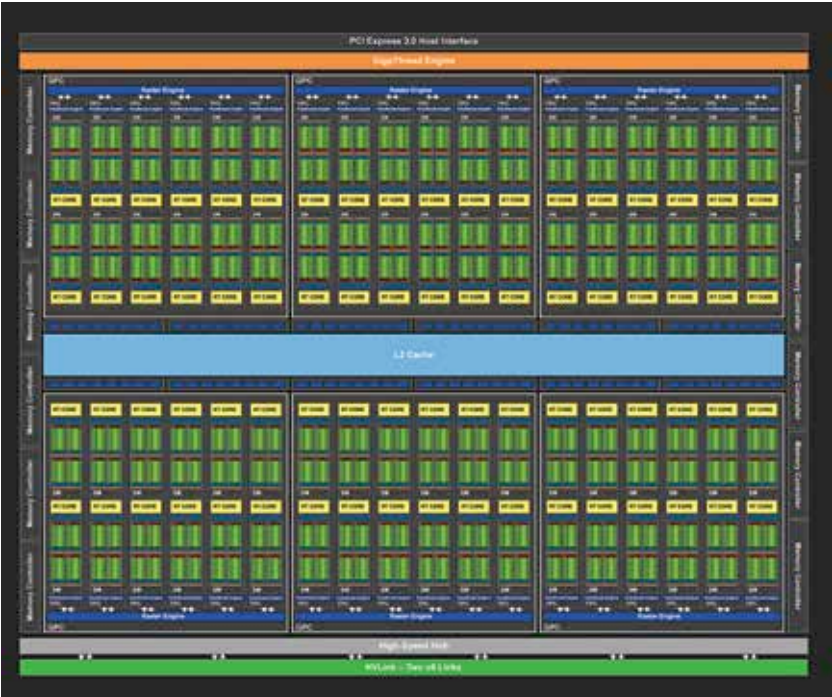
In a way, you could say that NVIDIA is reverting to the era of discrete shaders. In each SM, you had further bifurcation into four processing blocks with each consisting of 16 FP32 cores, 16 INT32 cores, 2 Tensor Cores, one warp scheduler and a dispatch unit. Each processing block has its own L0 instruction cache as well as a 64 KB register. Depending on the workload being handled (graphics or compute), the cache memory behaves differently.

One of the noteworthy changes with Turing was an additional data-path for non-Floating-Point operations. Previously, when a non-FP instruction was received, the FP-datapath would remain idle and thus, end up causing a delay in the pipeline. With a dedicated non-FP-datapath, the two types of instructions can be executed parallelly without holding up the pipeline.

As mentioned previously, there's the brand new real-time ray-tracing component which has modified the graphics pipeline in the new Turing cards. NVIDIA calls it the Hybrid Rendering Model. Ray-tracing isn't a new phenomenon, it's just that now NVIDIA has dedicated



NVIDIA's hybrid rendering pipeline in Turing



Turing TU102 Block diagram